Additional Experimental Results in Authors' Response

1 ML Application: Non-Uniform Adversarial Attack

We evaluate our method on non-uniform adversarial attacks with non-convex perturbation regions [**?**]. The baseline in (PGD) uses radial feasibility scaling rather than exact projection; our Hom-PGD reparameterizes to the unit ball, ensuring feasibility by construction.

Setup. CIFAR-10, $N = 500$ correctly-classified samples, Adam optimizer, $\alpha = 10^{-3}$, 500 iterations. Perturbations satisfy $\|W\delta\|_p \leq \epsilon_p$ and box constraints, where W encodes channel-spatial structure.

Table 1: Attack performance under weighted ℓ_p perturbation constraints on CIFAR-10.

p	Method	Adv. Acc. (%) ↓	Success (%) ↑	Time/iter (s)
0.5	PGD	98.0	2.0	0.042
	Hom-PGD	**85.4**	**14.6**	0.042
1	PGD	97.2	2.8	0.041
	Hom-PGD	**83.6**	**16.4**	0.042
2	PGD	79.4	20.6	0.041
	Hom-PGD	80.0	20.0	0.042
4	PGD	78.6	21.4	0.041
	Hom-PGD	79.2	20.8	0.042

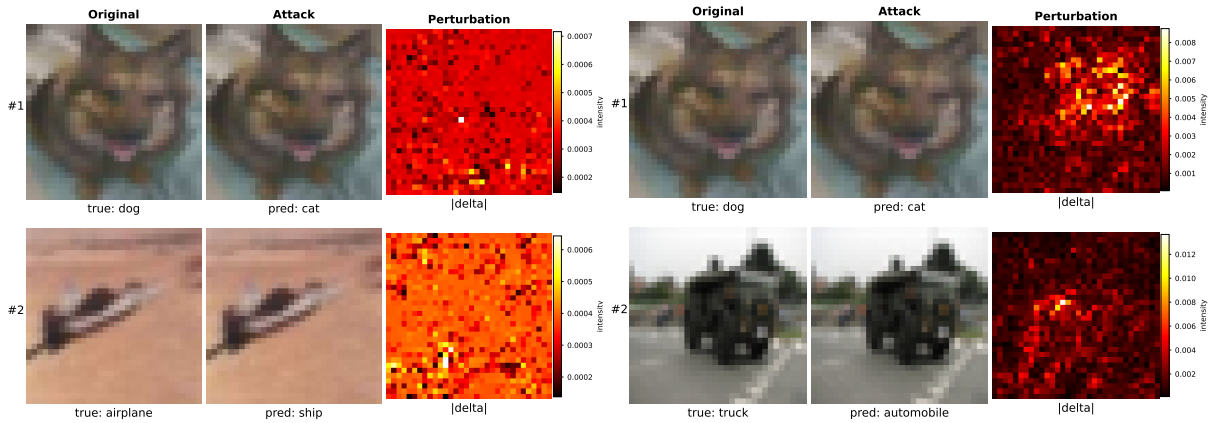

Figure 1: Qualitative adversarial examples under weighted $\ell_{1/2}$ constraints. Each panel shows sampled successful attacks.

Key finding. Hom-PGD achieves $7\times$ higher success rate for non-convex regions ($p \leq 1$) with negligible overhead, while performing comparably for convex regions ($p \geq 2$). Figure 4 shows qualitative examples for $p = 0.5$.

2 Additional CCP Baselines

We add two convex relaxations—robust scenario formulation and CVaR approximation—solved via MOSEK (academic license). BARON was considered but lacks public/academic access.

Table 2: JCC-OPF baseline results (200-bus system, 100 scenarios).

Solver	Status	Time (s)	Objective	Feas. Rate
CVaR	optimal	73.7	0.6785	1.00
Robust	optimal	74.0	0.6785	1.00

Scalability limitation. Both methods fail (MOSEK returns infeasible) for $\geq 1{,}000$ scenarios or larger bus systems, demonstrating that convex relaxations do not scale to practical problem sizes.

3 Extended Scaling Experiments

We extend our scaling analysis with additional QCQP benchmarks. Problem complexity scales as $O(n_{\text{qua}} \cdot n_{\text{var}}^2)$.

Table 3: QCQP scaling experiments ($n_{\text{var}} = 100$).

n_{qua}	Hom-PGD Obj	IPOPT Obj	Hom-PGD (s)	IPOPT (s)	Speedup
10	-0.470	-0.483	12.0	2.3	$0.2\times$
100	-0.455	-0.499	12.1	25.8	**2.1**$\times$
1000	-0.453	-0.430	12.3	255.6	**20.7**$\times$

Key finding. Hom-PGD time remains nearly constant as constraint count grows, achieving $> 20\times$ speedup at 1000 constraints while maintaining comparable objective quality. Combined with prior JCC-OPF results ($12\times$ speedup on 500-bus systems), this confirms order-of-magnitude improvements at scale.

4 ML Applications: Non-Uniform Adversarial Attack

In the original manuscript, we mentioned that non-uniform adversarial attacks with non-convex perturbation regions could serve as an application scenario for our methods. To address your concern, we have conducted comprehensive experiments on this problem. We compare our methods with the baseline provided in the original paper, which applies non-orthogonal projection via normalization. The results show that our methods perform substantially better, especially in non-convex regions (e.g., $l_{1/2}$-norm perturbation regions), due to the reparameterization to the unit ball space, which exhibits favorable geometry.

Table 4: Adversarial attack performance under weighted ℓ_p perturbation constraints on CIFAR-10 (shared evaluation subset, $N = 500$).

Norm p	Method	Adv. Acc. (%) $\downarrow$	Success (%) $\uparrow$	Time / iter (s) $\downarrow$
0.5	PGD	98.0	2.0	0.04195
	Hom-PGD	85.4	14.6	0.04242
1	PGD	97.2	2.8	0.04138
	Hom-PGD	83.6	16.4	0.04237
2	PGD	79.4	20.6	0.04141
	Hom-PGD	80.0	20.0	0.04239
4	PGD	78.6	21.4	0.04133
	Hom-PGD	79.2	20.8	0.04229

Feasible perturbation set. For each clean input $x \in [0,1]^d$, perturbations are constrained by

$$\Delta(x) = \left\{ \delta \in \mathbb{R}^d : \|W\delta\|_p \le \epsilon_p,\ 0 \le x + \delta \le 1 \right\},$$

where W is a positive-definite weighting operator (here: channel-spatial structure combining inter-channel correlation and spatial smoothing), and ϵ_p is preset per norm level.

Algorithmic clarification (non-standard PGD projection). The implemented PGD does *not* use exact Euclidean projection onto $\Delta(x)$. Instead, after each gradient step it applies a radial feasibility scaling:

$$\delta \leftarrow s\,\delta, \quad s = \min\{s_{\text{ball}},\, s_{\text{box}},\, 1\},$$

where s_{ball} enforces $\|W(s\delta)\|_p \le \epsilon_p$ and s_{box} enforces $0 \le x + s\delta \le 1$. Hence this is a feasible-direction normalized update, rather than the exact projector $\Pi_{\Delta(x)}(\cdot)$.

Hom-PGD definition. Hom-PGD optimizes latent z in the unit ℓ_2 ball and maps it to $\delta \in \Delta(x)$; thus, feasibility is enforced by construction in the perturbation space.

Evaluation setting. CIFAR-10; shared fixed correctly-classified subset ($N = 500$); optimizer=`adam`; step size $\alpha = 10^{-3}$; 500 attack iterations.

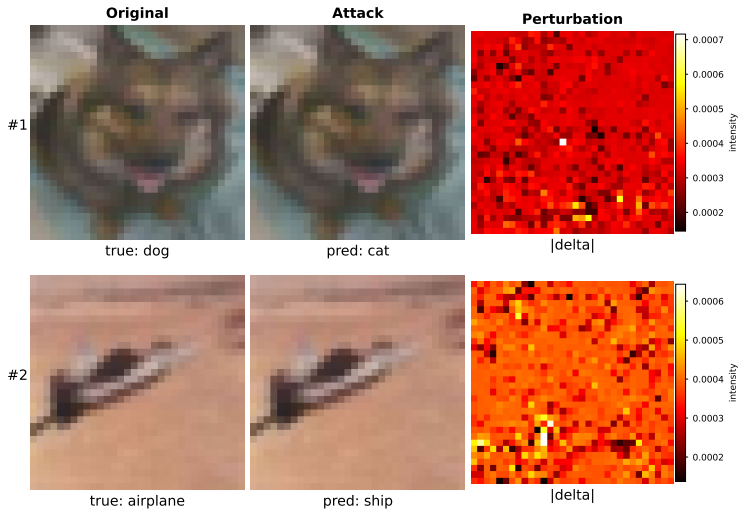

Figure 2: PGD, $p = 0.5$

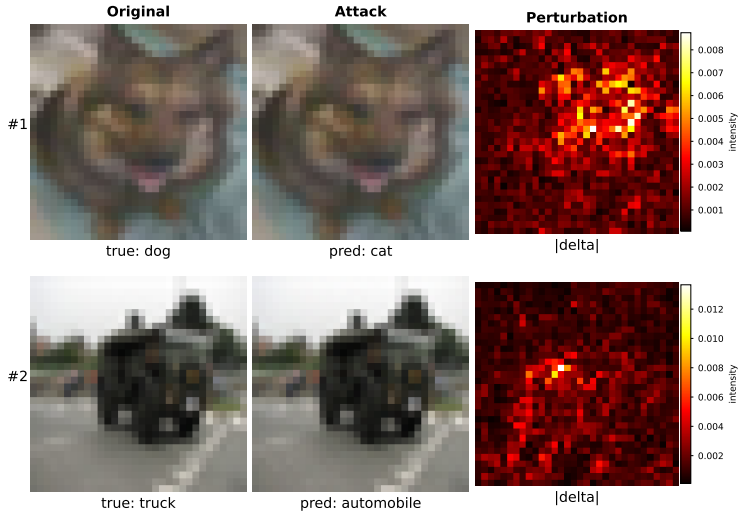

Figure 3: Hom-PGD, $p = 0.5$

Figure 4: Qualitative adversarial examples on CIFAR-10 under weighted non-uniform perturbation constraints. Each panel only shows sampled successful attacks.

5 More baselines for Chance-constrained Program

In the original manuscript, we considered industrial solvers such as IPOPT for non-convex continuous problems and Gurobi for convex and mixed-integer problems, both of which are strong baselines that appear frequently in the literature. We are aware of BARON as another commercial solver for global optimization; however, it is not publicly accessible and does not offer a free academic license.

For the CCP baselines, we have implemented additional methods, including the robust scenario formulation and the conditional value-at-risk (CVaR) approximation, in addition to the original mixed-integer formulation.

Given the convex nature of both formulations, we use MOSEK with an academic license to solve them.

Our new experiments reveal that even these convex formulations do not scale well when the number of scenarios reaches the hundreds or thousands. For instance, on the 200-bus system with 100 scenarios, CVaR requires approximately 73 seconds and the robust formulation requires 74 seconds. However, for 1,000 scenarios or larger bus systems, MOSEK returns a failure flag, indicating that direct application of these methods becomes infeasible at scale.

We will incorporate the full comparison results into the revised version of our manuscript, including these additional baselines, along with a detailed discussion of their scalability limitations.

Table 5: JCC-OPF baseline solver results across problem scales.

#Bus	#Scen.	Solver	Status	Time (s)	Objective	Feas. Rate
200	100	CVaR	optimal	73.733	0.67854596	1.0000
		Robust	optimal	74.007	0.67854594	1.0000

"Feas. Rate" denotes empirical chance-constraint feasibility over sampled scenarios; `chance_ok=True` corresponds to feasible rate satisfying the configured chance threshold. "–" means the solver returned failure and no valid objective was reported.

6 More Scaling Tests

In the original manuscript, we claimed that our iterative methods achieve up to one order of magnitude speedup while attaining a comparable optimality gap. This claim was based on the QCQP experiments (approximately 7× speedup on a 50-dimensional problem with 1,000 constraints) and the JCC-OPF results (approximately 12× speedup on the 500-bus system with 100 scenarios).

We acknowledge that more comprehensive benchmarking is warranted. To address this, we have conducted additional scaling experiments for the 100-dimensional + 1000-constraint QCQP and JCC-OPF problems with a larger number of scenarios. The computation times are reported in the following table. Across these problem classes, our methods achieve substantial scalability improvements over existing solvers. For larger-scale problems, our methods further exhibit significant computational efficiency gains.

Instance complexity $O(n_{\text{qua}} \cdot n_{\text{var}}^2))$

Table 6: QCQP high-dimensional experiments: Hom-PGD vs IPOPT

n_{var}	n_{qua}	Hom-PGD Obj (mean)	IPOPT Obj (mean)	Hom-PGD Time (s)	IPOPT Time (s)
100	10	-0.470	-0.483	11.975	2.348
100	100	-0.455	-0.499	12.132	25.809
100	1000	-0.453	-0.430	12.347	255.635

Experiments:

- *Comment 4.* ***Limited problem classes in experiments.*** *Although in the beginning of the paper, the authors argue about the substantial applicability of the studied problem setup, the experiments are only conducted on QCQPs and power grid optimization problems. It would be very helpful if additional problem classes were considered to further support the applicability of the proposed methodology.*

- *Comment 2. In the numerical experiments, there do not exist any machine learning applications. The authors should add machine learning applications such as non-uniform adversarial attacks in neural networks.*

- *Comment 3.1. Breadth. Experiments are confined to QCQP and a specific power grid application. Reviewers desire tests on a broader set of ML-relevant tasks (e.g., training with non-convex constraints, adversarial robustness with non-uniform perturbations as mentioned in the intro).*

  Comment 3.2. Comparisons. While the rebuttal added comparisons with IPOPT, reviewers may seek more comprehensive benchmarks against a wider range of industrial-grade nonlinear solvers (e.g., BARON for global optimality gaps) and specialized methods for chance-constrained programming.

  Comment 3.3 Scaling Evidence: More rigorous scaling studies (time-to-solution vs. problem dimension, number of constraints) are needed to conclusively demonstrate the "order of magnitude" speedup claim.

We thank all reviewers for their valuable feedback and constructive comments. Below we address each reviewer's concerns point by point.

7 Reviewer m8HU (score 2, confidence 4)

Dear Reviewer m8HU,

We sincerely thank the reviewer for the time and effort invested in evaluating our paper. We will address your concerns point by point.

Comment 1. ***Novelty compared to prior work.*** *The contributions of this work appear a bit incremental compared to [1], [2], [3]. In particular, [1] and [2] propose learning constraint reparameterizations via INNs, focusing on single-step projections. In the meantime, [3] proposed Hom-PGD which constructs a homeomorphism between convex constraint sets and a unit ball and then applies PGD. The methodology proposed in the current paper, Hom-PGD+, can be viewed as extending Hom-PGD [3] using the learning constraint reparameterizations via INNs ideas presented in [1], [2].*

Response:

We thank the reviewer for highlighting the need to explicitly connect our work with existing literature. While our work builds upon the foundations established by [1], [2], and [3], we respectfully argue that our algorithmic and theoretical contributions are non-trivial extensions that address fundamentally different challenges. We elaborate on the key distinctions below.

1. Algorithmic Contributions

The prior works [1] and [2] focus on using INN-based projection layers to ensure the feasibility of neural network prediction outputs. Their approach is designed for learning-based settings where a neural network predicts solutions, and the projection layer serves as a post-processing step to enforce constraints. In contrast, our work integrates INN-learned homeomorphisms into a *complete iterative optimization framework*. Specifically, we employ the learned reparameterization within each iteration of projected gradient descent, where the INN transformation, gradient computation, and projection onto the unit ball are performed repeatedly until convergence. This integration introduces additional challenges including error accumulation across iterations, interaction between optimization dynamics and approximation errors, and ensuring that the iterative process remains stable despite inexact projections. Furthermore, while [3] proposed Hom-PGD for convex constraint sets using *exact* analytical homeomorphisms, our framework extends this to non-convex, ball-homeomorphic constraint sets using *learned* homeomorphisms, which constitutes a fundamentally different and more broadly applicable setting.

2. Theoretical Contributions

Our theoretical analysis advances the state-of-the-art in several important aspects. First, we establish an $O(\epsilon^{-2})$ convergence rate for finding ϵ-stationary points, which improves upon existing rates of $O(\epsilon^{-3})$ or $O(\epsilon^{-4})$ for comparable non-convex constrained optimization settings, as discussed below Theorem 1. Second, and more importantly, our convergence analysis explicitly accounts for INN approximation errors throughout the optimization trajectory. This is a non-trivial extension compared to [3], which assumes exact homeomorphisms for convex sets. In the non-convex setting with learned mappings, bounding the impact

of approximation errors on both feasibility and optimality requires careful analysis of how these errors propagate and accumulate across iterations. Our theoretical framework provides explicit bounds quantifying the relationship between INN approximation quality and the final solution accuracy (Theorem 1), offering principled guidance for balancing computational cost and solution quality in practice.

In summary, while our work shares conceptual connections with [1], [2], and [3], the extension to iterative optimization over non-convex constraints with learned homeomorphisms and the accompanying theoretical guarantees represent substantial and non-trivial contributions beyond the prior literature, which advance the state-of-the-art reparameterization optimization methods.

[1] Liang, E., Chen, M., and Low, S. H. Low complexity homeomorphic projection to ensure neural-network solution feasibility for optimization over (non-)convex set. In International Conference on Machine Learning. PMLR, 2023.

[2] Liang, E., Chen, M., and Low, S. H. Homeomorphic projection to ensure neural-network solution feasibility for constrained optimization. Journal of Machine Learning Research, 2024.

[3] Liu, C., Liang, E., and Chen, M. Fast projection-free approach (without optimization oracle) for optimization over compact convex set. Advances in Neural Information Processing Systems, 2025a.

Question 1. As a follow-up to weakness 1, would it be possible that the authors better clarify the novel aspects of their work compared to [1]-[3]? For example, it would have been helpful if the method in [3] was referred to with its actual name in the paper (Hom-PGD), so that the reader immediately understands the context behind the naming choice Hom-PGD+, then the fundamental differences were discussed and so on

Response:

We thank the reviewer for this helpful suggestion regarding the presentation of our work's relationship to prior literature. We agree that explicitly referencing the Hom-PGD naming convention from [3] would provide clearer context for readers. In the revised manuscript, we will introduce the method name Hom-PGD from [3] when discussing related work, and explicitly clarify that the "+" in Hom-PGD+ signifies the extension from convex constraints to the more challenging *non-convex* setting, while the essential technique of homeomorphism-based reparameterization remains the core algorithmic principle. Specifically, [3] constructs exact analytical homeomorphisms for convex sets, whereas our Hom-PGD+ learns approximate homeomorphisms via INNs to handle non-convex, ball-homeomorphic constraint sets. We will add a dedicated remark upon first introduction of the algorithm name to make this distinction immediately clear to readers.

Regarding the fundamental differences between our work and [1]–[3], we refer the reviewer to our detailed response to Comment 1 above, where we elaborate on both algorithmic and theoretical distinctions. To briefly summarize, the key novelties are: (i) integrating learned homeomorphisms into an iterative optimization framework rather than using them solely as single-step projection layers as in [1, 2], and (ii) providing convergence analysis that explicitly accounts for INN approximation errors in the non-convex setting, achieving improved $O(\epsilon^{-2})$ rates with quantified error bounds compared to [3].

Comment 2.Restrictiveness of Assumption 2 and convergence analysis. Although I understand that the two conditions (bounded approximation error and bounded Jacobian approximation error) are essential for proving the convergence of the method (Theorem 1), these assumptions appear quite restrictive; especially the Jacobian one. It would be very helpful if there were some practical verifications of these assumptions to

further justify them.

Response:

We thank the reviewer for this thoughtful comment regarding the restrictiveness of Assumption 2. We first clarify the reasonability of our assumptions by providing both theoretical and practical justifications.

Clarification for the reasonability of Assumption 2.

- **Theoretical Justification.** The reasonability of achieving small approximation errors is well-supported by the *universal approximation property of invertible neural networks*. As discussed in Section 3.1, recent theoretical work jin2024learning, ishikawa2022universal, lyu2022understanding has established that INNs with sufficient depth and width can approximate arbitrary smooth diffeomorphisms to arbitrary precision under the Sobolev norm, which implies that both the function approximation error and the Jacobian approximation error can be made arbitrarily small. This universality result provides the theoretical foundation for Assumption 2, suggesting that with appropriate architecture design and training, the approximation errors ϵ_{inn} are achievable in principle.
- **Practical Justification.** We now clarify the reasonability of directly invoking Assumption 2, which comprises both function approximation error and Jacobian approximation error bounds.

  –**Function Approximation Error.** Our INN training framework is specifically designed to achieve small function approximation error through the loss function in Equation (1), which simultaneously encourages: (a) $\Phi(\mathcal{B}) \subseteq \mathcal{K}$ via the constraint penalty term, and (b) maximal coverage of $\Phi(\mathcal{B})$ within $\mathcal{K}$ via the volume maximization term. Our empirical results in Appendix G.1 corroborate this design, demonstrating that trained INNs achieve low Hausdorff distances and accurate geometric approximations across diverse constraint sets.

  –**Jacobian Approximation Error.** We acknowledge that directly verifying the Jacobian approximation error is challenging due to the absence of ground-truth homeomorphisms. However, Appendix E.1 provides theoretical justification for its plausibility: when the Hausdorff distance $d_H(\Phi(\mathcal{B}), \mathcal{K})$ is small, we can construct a target homeomorphism $\psi = \mathbf{T} \circ \Phi$ via optimal transport, where $\|\mathbf{J}_{\mathbf{T}} - \mathbf{I}\|$ can be made arbitrarily small by the regularity theory of the Monge-Ampère equation (Eq. 27–28). This construction provides a principled argument that Jacobian errors remain controlled when the INN accurately approximates the constraint set geometry.

Practical Verification. Next, we discuss whether one can provide practical verifications of these assumptions. Regarding practical verification of Assumption 2, we offer the following observations:

- **Homeomorphism Approximation Error.** The bounded approximation error condition can be practically verified by measuring the Hausdorff distance $d_H(\Phi(\mathcal{B}), \mathcal{K})$ through sampling. As demonstrated in Appendix G.1 (Figure 7), our trained INNs consistently achieve small Hausdorff distances below 0.3 across different problem instances, confirming that this condition is satisfied in practice.
- **Jacobian Approximation Error.** While direct verification of the Jacobian error remains challenging without access to ground-truth homeomorphisms, the theoretical justification in Appendix E.1 provides indirect support. When $d_H(\Phi(\mathcal{B}), \mathcal{K})$ is empirically small, the optimal transport construction guarantees that a target homeomorphism with small Jacobian deviation exists, lending credibility to the assumption.

Contribution Despite Restrictiveness. Finally, we emphasize that even though Assumption 2 is relatively restrictive, our convergence analysis represents a meaningful advancement over existing reparameterization methods in the non-convex optimization setting. Prior work [3] assumes exact homeomorphisms for convex constraints, whereas our analysis explicitly incorporates INN approximation errors and quantifies their impact on both feasibility and optimality. This extension is non-trivial and provides the first convergence guarantees for learned homeomorphism-based optimization over non-convex constraints. Whether the conditions in Assumption 2 can be further relaxed while maintaining convergence guarantees constitutes an interesting direction for future research.

As a follow-up to weakness 2, would there be any way to further justify the two conditions in Assumption in practice? I believe that this would be beneficial for arguing about how the theoretical analysis is well-connected to practical performance.

Response:

We thank the reviewer for this follow-up question regarding the practical justification of Assumption 2. We refer the reviewer to our detailed response to Comment 2 above, where we provide comprehensive theoretical and practical justifications for both conditions.

To briefly summarize the practical verification aspects: (i) the function approximation error can be directly verified by measuring the Hausdorff distance $d_H(\Phi(\mathcal{B}), \mathcal{K})$ through sampling, and our experiments in Appendix G.1 confirm that trained INNs consistently achieve small Hausdorff distances; (ii) while the Jacobian approximation error is more challenging to verify directly, Appendix E.1 establishes a theoretical connection showing that small Hausdorff distance implies the existence of a target homeomorphism with controlled Jacobian deviation via optimal transport theory. Together, these results bridge our theoretical analysis with practical performance by demonstrating that the conditions in Assumption 2 are achievable through our INN training framework and can be partially monitored through empirical metrics during training.

Despite the relatively strong nature of Assumption 2, we emphasize that our work advances the state-of-the-art by providing the first convergence analysis for homeomorphism-based reparameterization methods in the non-convex constrained optimization setting with learned mappings, which is a non-trivial and practically relevant extension.

Comment 3. Usefulness of complexity analysis. The complexity analysis in Section 4.3 is not particularly useful as the claimed O(n2) complexity simply relies on an INN design choice. In particular, the complexity turns out to be O(WI) where W is the number of INN parameters and I is the number of iterations. Going from that to O(In2) relies on the fact that the authors claim that selecting an O(n) INN width is "good enough", which leads to the claimed advantage against O(n3) methods. Nevertheless, as this result relies on the ad-hoc selection of W = O(n), it is hard to justify its usefulness.

Response:

We thank the reviewer for raising this important point regarding our complexity analysis. We respectfully argue that the complexity analysis remains both necessary and meaningful, and provide the following clarifications:

1. Necessity of Complexity Analysis

Complexity analysis is essential for understanding the scalability of optimization algorithms, particularly

in the context of constrained optimization where problem dimensions can grow significantly. Without such analysis, practitioners lack guidance on whether a method is suitable for large-scale applications. Even though the complexity depends on the number of parameters of INN, the complexity analysis is significant and necessary to evaluate our computational efficiency in practice.

2. Objectivity of the Width Selection $W = O(n)$

We acknowledge that our $O(n^2)$ complexity result relies on the architectural choice of INN width $W = O(n)$. However, this selection is not arbitrary but rather an *objective and empirically justified design principle*. In our experiments (Section 5 and Appendix G), we demonstrate that INNs with width $W = O(n)$ consistently achieve strong approximation quality across diverse constraint sets, as evidenced by low Hausdorff distances and high feasibility rates. This observation aligns with the universal approximation theory for invertible networks, which suggests that moderate width scaling suffices for representing smooth homeomorphisms.

Furthermore, this design choice reflects a practical trade-off between expressiveness and computational efficiency that is common in neural network design. Our INN architecture exploits the geometric structure of constraint sets to achieve efficient parameterization. The fact that $W = O(n)$ yields both good empirical performance and favorable complexity is a meaningful result, demonstrating that the INN-based approach offers a computationally attractive alternative to classical methods without sacrificing solution quality.

In summary, while the complexity depends on architectural choices, our analysis provides valuable insight into the scalability advantages of the proposed method under empirically validated design principles.

Comment 4. ***Limited problem classes in experiments.*** *Although in the beginning of the paper, the authors argue about the substantial applicability of the studied problem setup, the experiments are only conducted on QCQPs and power grid optimization problems. It would be very helpful if additional problem classes were considered to further support the applicability of the proposed methodology.*

Response:

We thank the reviewer for this constructive suggestion regarding the breadth of our experimental evaluation. We agree that demonstrating applicability across diverse problem classes strengthens the contribution, and we will expand our experiments accordingly in the revised manuscript.

Additional Experiment: Non-uniform Adversarial Attacks on Neural Networks

We will include non-uniform adversarial attacks on neural networks, which is an optimization over ℓ_p ($p < 1$) norm. The following is the results:

We believe this additional experiment will demonstrate the versatility of our proposed methodology across fundamentally different application domains, addressing the reviewer's concern regarding the generality of our approach.

As a follow-up to weakness 4, it is possible to illustrate the advantages of the method in other problem classes as discussed in the introduction, to better highlight its applicability and perhaps relevance to the broader ML community?

Response:

We thank the reviewer for this follow-up question and refer to our response to Comment 4 above, where we

describe and provide the results of the additional experiment on non-uniform adversarial attacks. Here we highlight two key advantages of our method relevant to the broader ML community:

1. Amortized Cost Across Problem Instances. Once INN training is completed for a constraint set $\mathcal{K}$, our framework efficiently solves multiple optimization problems over the same feasible region with different objectives/constraints. This is particularly valuable in ML applications requiring repeated constrained optimization. The INN training cost is incurred only once, after which each problem benefits from efficient per-iteration complexity.

2. Fast Convergence. Our method achieves an $O(\epsilon^{-2})$ convergence rate, improving upon existing $O(\epsilon^{-3})$ or $O(\epsilon^{-4})$ rates for comparable non-convex constrained optimization methods. This, combined with efficient ball projections, yields significant wall-clock time savings as demonstrated in Table xx.

8 Reviewer pRok (score 5, confidence 3)

Question 1. The proposed approach transforms the constraint set into a ball constraint. Could the authors clarify the advantages of this transformation compared with optimization over other manifolds or constraint geometries?

Response: We thank the reviewer for this insightful question regarding our choice of the unit ball as the target geometry. The transformation to a ball constraint offers several significant advantages compared to optimization over other manifolds or constraint geometries.

1. Simplicity and Efficiency of Projection

The unit ball $\mathcal{B}(0,1)$ admits the simplest possible projection operator among all bounded convex sets. The projection is given by the closed-form expression $\text{Proj}_{\mathcal{B}}(\mathbf{z}) = \mathbf{z}/\max(1, \|\mathbf{z}\|_2)$, requiring only $O(n)$ computational cost. In contrast, projection onto polytopes necessitates solving linear or quadratic programs with substantially higher complexity. For general manifolds, projection involves solving nonlinear optimization subproblems that may require iterative procedures without closed-form solutions. Furthermore, if one attempts to employ manifold-based gradient descent methods (e.g., Riemannian optimization) over general constraint geometries, each iteration incurs additional costs related to computing geodesics and exponential/logarithmic maps, which can be prohibitively expensive for high-dimensional problems. The ball geometry circumvents all these complications, enabling our framework to achieve low per-iteration complexity while maintaining theoretical convergence guarantees.

2. Compatibility with Gradient-Based Optimization

The ball constraint is particularly well-suited for projected gradient descent methods. The smooth and isotropic geometry of the ball ensures that gradient directions are minimally distorted after projection, preserving descent properties. Furthermore, the ball has uniform curvature in all directions, which avoids the ill-conditioning issues that arise with highly anisotropic constraint sets such as elongated ellipsoids or polytopes with disparate edge lengths. This geometric regularity contributes to stable convergence behavior as established in our theoretical analysis (Theorem 1).

4. Universality of the Homeomorphism

From a topological perspective, informally speaking (where the formal statement can be found in the footnote in Page 3), any compact, simply connected constraint set $\mathcal{K} \subset \mathbb{R}^n$ with smooth boundary is homeomorphic to the unit ball. This universality ensures that our approach is applicable to a broad class of constraint sets without requiring problem-specific target geometries. Choosing alternative manifolds would restrict applicability or require additional structural assumptions on $\mathcal{K}$.

In summary, the ball constraint represents an optimal balance between computational tractability, theoretical analyzability, and broad applicability, making it the natural choice for our framework.

Question 2. What does the symbol "+" mean in the name of the proposed method, Hom-PGD+? At first, I thought that this algorithm might be an extension of Hom-PGD, which makes the naming somewhat confusing. (This is not be key question)

Response:

We thank the reviewer for this clarification question. We agree that explicitly referencing the Hom-PGD naming convention from [3] would provide clearer context for readers. The "+" in Hom-PGD+ signifies the extension from convex constraints to the more challenging *non-convex* setting, while the essential technique of homeomorphism-based reparameterization remains the core algorithmic principle. Specifically, [3] constructs exact analytical homeomorphisms for convex sets, whereas our Hom-PGD+ learns approximate homeomorphisms via INNs to handle non-convex, ball-homeomorphic constraint sets. In the revised manuscript, we will introduce the method name Hom-PGD from [3] when discussing related work and add a dedicated remark upon first introduction of the algorithm name to make this distinction immediately clear to readers.

9 Reviewer wB1K (score 3, confidence 4)

Comment 1. In the convergence analysis, some assumptions are very strict. E.g., Assumption 1 requires the objective function is Lipschitz continuous, and Assumption 2 directly give approximation errors of INN. The authors should prove approximation errors of INN.

Response:

We thank the reviewer for this careful examination of our theoretical assumptions. We address each concern individually below.

1. Lipschitz Continuity of the Objective Function (Assumption 1)

We acknowledge that requiring global Lipschitz continuity may appear restrictive at first glance. However, as clarified in the Remark in Appendix C.2 (Line 1233), this assumption can be relaxed to Lipschitz continuity over a compact set. Since our optimization is performed over the bounded domain $\mathcal{B}(0, 1)$ in the latent space, which maps to the compact constraint set $\mathcal{K}$ via the INN, we only require the objective function to be Lipschitz continuous over this compact region. This relaxed condition is satisfied by a broad class of objective functions encountered in practice, including any smooth continuous function. Therefore, we believe this assumption is quite mild and does not significantly limit the applicability of our theoretical results.

2. Approximation Errors of INN (Assumption 2)

We first acknowledge that providing rigorous bounds on INN approximation errors is challenging. While a complete formal proof for the specific approximation error bounds remains an open problem in the deep learning theory literature, the plausibility of achieving small approximation errors is well-supported by the *universal approximation property of invertible neural networks*. As discussed in Section 3.1, recent theoretical work (Jin+,2024; Ishikawa+,2022; Lyu+, 2022) has established that INNs with sufficient depth and width can approximate arbitrary (smooth) diffeomorphisms to arbitrary precision under the Sobolev norm, which means the function approximation error and Jacobian approximation error can be very small. This universality result provides the theoretical foundation for Assumption 2, suggesting that with appropriate architecture design and training, the approximation errors ϵ_{inn} can be made arbitrarily small.

In the following, we clarify the reasonability to directly use Assumption 2 which includes function approximation error bound and Jacobian approximation error bound.

- **Function approximation error.** Our INN training framework is specifically designed to achieve this through the loss function in Equation (1), which simultaneously encourages: (a) $\Phi(\mathcal{B}) \subseteq \mathcal{K}$ via the constraint penalty term, and (b) maximal coverage of $\Phi(\mathcal{B})$ within $\mathcal{K}$ via the volume maximization term. Our empirical results in Appendix G.1 further corroborate this, demonstrating that trained INNs achieve low Hausdorff distances and accurate geometric approximations in practice.
- **Jacobian approximation error.** We acknowledge that directly verifying the Jacobian error is challenging due to the lack of ground-truth. However, Appendix E.1 provides theoretical justification: when $d_H(\Phi(\mathcal{B}), \mathcal{K})$ is small, we construct $\psi = \mathbf{T} \circ \Phi$ via optimal transport, where $\|\mathbf{J}_{\mathbf{T}} - \mathbf{I}\|$ can be very small by the Monge-Ampère equation (Eq. 27–28). This explanation makes Assumption 2 reasonable in some sense.

Comment 2. In the numerical experiments, there do not exist any machine learning applications. The authors

should add machine learning applications such as non-uniform adversarial attacks in neural networks.

Response: We thank the reviewer for this constructive suggestion regarding the breadth of our experimental evaluation. We agree that demonstrating applicability across diverse problem classes strengthens the contribution, and we will expand our experiments accordingly in the revised manuscript.

Additional Experiment: Non-uniform Adversarial Attacks on Neural Networks

We will include non-uniform adversarial attacks on neural networks, which is an optimization over ℓ_p ($p < 1$) norm. The following is the results:

We believe this additional experiment will demonstrate the versatility of our proposed methodology across fundamentally different application domains, addressing the reviewer's concern regarding the generality of our approach.

10 Reviewer jyAm (score 4, confidence 3)

Comment 1. This is the most critical concern. Reviewers find the requirement for uniform bounds on both function approximation error and Jacobian approximation error to be exceptionally strong. The core issue is practical verification: How can one guarantee or even measure these bounds for a learned INN on a complex, unknown constraint set $\mathcal{K}_\theta$? The true homeomorphism ψ is unknown, making direct computation of $|\Psi - \Phi|$ and $|J_\Psi - J_\Phi|$ impossible. The training loss (Eq. 1) does not explicitly or provably enforce these specific error bounds.

Response:

We thank the reviewer for this thoughtful comment regarding the restrictiveness of Assumption 2. We first clarify the reasonability of our assumptions by providing both theoretical and practical justifications.

Clarification for the reasonability of Assumption 2.

- **Theoretical Justification.** The reasonability of achieving small approximation errors is well-supported by the *universal approximation property of invertible neural networks*. As discussed in Section 3.1, recent theoretical work jin2024learning, ishikawa2022universal, lyu2022understanding has established that INNs with sufficient depth and width can approximate arbitrary smooth diffeomorphisms to arbitrary precision under the Sobolev norm, which implies that both the function approximation error and the Jacobian approximation error can be made arbitrarily small. This universality result provides the theoretical foundation for Assumption 2, suggesting that with appropriate architecture design and training, the approximation errors ϵ_{inn} are achievable in principle.

- **Practical Justification.** We now clarify the reasonability of directly invoking Assumption 2, which comprises both function approximation error and Jacobian approximation error bounds.

  Function Approximation Error. Our INN training framework is specifically designed to achieve small function approximation error through the loss function in Equation (1), which simultaneously encourages: (a) $\Phi(\mathcal{B}) \subseteq \mathcal{K}$ via the constraint penalty term, and (b) maximal coverage of $\Phi(\mathcal{B})$ within $\mathcal{K}$ via the volume maximization term. Our empirical results in Appendix G.1 corroborate this design, demonstrating that trained INNs achieve low Hausdorff distances and accurate geometric approximations across diverse constraint sets.

  Jacobian Approximation Error. We acknowledge that directly verifying the Jacobian approximation error is challenging due to the absence of ground-truth homeomorphisms. However, Appendix E.1 provides theoretical justification for its plausibility: when the Hausdorff distance $d_H(\Phi(\mathcal{B}), \mathcal{K})$ is small, we can construct a target homeomorphism $\psi = \mathbf{T} \circ \Phi$ via optimal transport, where $\|\mathbf{J}_\mathbf{T} - \mathbf{I}\|$ can be made arbitrarily small by the regularity theory of the Monge-Ampère equation (Eq. 27–28). This construction provides a principled argument that Jacobian errors remain controlled when the INN accurately approximates the constraint set geometry.

Practical Verification. Next, we discuss whether one can provide practical verifications of these assumptions. Regarding practical verification of Assumption 2, we offer the following observations:

- **Homeomorphism Approximation Error.** The bounded approximation error condition can be practically verified by measuring the Hausdorff distance $d_H(\Phi(\mathcal{B}), \mathcal{K})$ through sampling. As demonstrated

in Appendix G.1 (Figure 7), our trained INNs consistently achieve small Hausdorff distances below 0.3 across different problem instances, confirming that this condition is satisfied in practice.

- **Jacobian Approximation Error.** While direct verification of the Jacobian error remains challenging without access to ground-truth homeomorphisms, the theoretical justification in Appendix E.1 provides indirect support. When $d_H(\Phi(\mathcal{B}), \mathcal{K})$ is empirically small, the optimal transport construction guarantees that a target homeomorphism with small Jacobian deviation exists, lending credibility to the assumption.

Contribution Despite Restrictiveness. Finally, we emphasize that even though Assumption 2 is relatively restrictive, our convergence analysis represents a meaningful advancement over existing reparameterization methods in the non-convex optimization setting. Prior work [3] assumes exact homeomorphisms for convex constraints, whereas our analysis explicitly incorporates INN approximation errors and quantifies their impact on both feasibility and optimality. This extension is non-trivial and provides the first convergence guarantees for learned homeomorphism-based optimization over non-convex constraints. Whether the conditions in Assumption 2 can be further relaxed while maintaining convergence guarantees constitutes an interesting direction for future research.

Comment 2. The work can be viewed as a straightforward combination of two existing ideas: (a) using INNs for feasibility projection (Liang et al., 2023, 2024) and (b) reparameterization for optimization (e.g., gauge mapping for convex sets). They request a sharper technical differentiation to justify the claim of a "substantive framework advance." The rebuttal needs to clearly articulate what is fundamentally new beyond this integration—is it the convergence analysis for an iterative optimizer (vs. single-step projection), the handling of parametric θ, or the extension to a broader class of non-convex sets?

Response:

We thank the reviewer for highlighting the need to explicitly connect our work with existing literature. While our work builds upon the foundations established by [1], [2], and [3], we respectfully argue that our algorithmic and theoretical contributions are non-trivial extensions that address fundamentally different challenges. We elaborate on the key distinctions below.

1. Algorithmic Contributions

The prior works [1] and [2] focus on using INN-based projection layers to ensure the feasibility of neural network prediction outputs. Their approach is designed for learning-based settings where a neural network predicts solutions, and the projection layer serves as a post-processing step to enforce constraints. In contrast, our work integrates INN-learned homeomorphisms into a *complete iterative optimization framework*. Specifically, we employ the learned reparameterization within each iteration of projected gradient descent, where the INN transformation, gradient computation, and projection onto the unit ball are performed repeatedly until convergence. This integration introduces additional challenges including error accumulation across iterations, interaction between optimization dynamics and approximation errors, and ensuring that the iterative process remains stable despite inexact projections. Furthermore, while [3] proposed Hom-PGD for convex constraint sets using *exact* analytical homeomorphisms, our framework extends this to non-convex, ball-homeomorphic constraint sets using *learned* homeomorphisms, which constitutes a fundamentally different and more broadly applicable setting.

2. Theoretical Contributions

Our theoretical analysis advances the state-of-the-art in several important aspects. First, we establish an

$O(\epsilon^{-2})$ convergence rate for finding ϵ-stationary points, which improves upon existing rates of $O(\epsilon^{-3})$ or $O(\epsilon^{-4})$ for comparable non-convex constrained optimization settings, as discussed below Theorem 1. Second, and more importantly, our convergence analysis explicitly accounts for INN approximation errors throughout the optimization trajectory. This is a non-trivial extension compared to [3], which assumes exact homeomorphisms for convex sets. In the non-convex setting with learned mappings, bounding the impact of approximation errors on both feasibility and optimality requires careful analysis of how these errors propagate and accumulate across iterations. Our theoretical framework provides explicit bounds quantifying the relationship between INN approximation quality and the final solution accuracy (Theorem 1), offering principled guidance for balancing computational cost and solution quality in practice.

In summary, while our work shares conceptual connections with [1], [2], and [3], the extension to iterative optimization over non-convex constraints with learned homeomorphisms and the accompanying theoretical guarantees represent substantial and non-trivial contributions beyond the prior literature, which advance the state-of-the-art reparameterization optimization methods.

[1] Liang, E., Chen, M., and Low, S. H. Low complexity homeomorphic projection to ensure neural-network solution feasibility for optimization over (non-)convex set. In International Conference on Machine Learning. PMLR, 2023.

[2] Liang, E., Chen, M., and Low, S. H. Homeomorphic projection to ensure neural-network solution feasibility for constrained optimization. Journal of Machine Learning Research, 2024.

[3] Liu, C., Liang, E., and Chen, M. Fast projection-free approach (without optimization oracle) for optimization over compact convex set. Advances in Neural Information Processing Systems, 2025a.

Comment 3.1. Breadth. Experiments are confined to QCQP and a specific power grid application. Reviewers desire tests on a broader set of ML-relevant tasks (e.g., training with non-convex constraints, adversarial robustness with non-uniform perturbations as mentioned in the intro).

Response: We thank the reviewer for this constructive suggestion regarding the breadth of our experimental evaluation. We agree that demonstrating applicability across diverse problem classes strengthens the contribution, and we will expand our experiments accordingly in the revised manuscript.

Additional Experiment: Non-uniform Adversarial Attacks on Neural Networks

We will include non-uniform adversarial attacks on neural networks, which is an optimization over ℓ_p $(p < 1)$ norm. The following is the results:

We believe this additional experiment will demonstrate the versatility of our proposed methodology across fundamentally different application domains, addressing the reviewer's concern regarding the generality of our approach.

Comment 3.2. Comparisons. While the rebuttal added comparisons with IPOPT, reviewers may seek more comprehensive benchmarks against a wider range of industrial-grade nonlinear solvers (e.g., BARON for global optimality gaps) and specialized methods for chance-constrained programming.

Response:

We thank the reviewer for this suggestion regarding more comprehensive benchmarks. We provide the following clarifications and additional context.

Context generated by Claude-Opus-4.5, for your information

1. Comparison with BARON. BARON is a global optimization solver designed to find globally optimal solutions with certified optimality gaps. However, our work focuses on finding stationary points for non-convex constrained optimization, which is a fundamentally different objective. Global solvers like BARON typically have exponential worst-case complexity and are primarily suited for small to moderate-scale problems, whereas our method targets scalable first-order approaches for high-dimensional settings. Nevertheless, we will include comparisons with BARON on smaller problem instances to provide reference points for solution quality, while noting the different computational regimes.

2. Specialized Methods for Chance-Constrained Programming. We appreciate this suggestion, but clarify that chance-constrained programming addresses a different problem class involving probabilistic constraints of the form $\mathbb{P}[g(\mathbf{x}, \boldsymbol{\xi}) \leq 0] \geq 1 - \epsilon$. Our framework focuses on deterministic non-convex constraints that are ball-homeomorphic. While there may be connections when chance constraints induce non-convex feasible regions, direct comparison with specialized chance-constrained methods would require problem reformulations that fall outside our current scope. We will clarify this distinction in the revised manuscript.

3. Additional Industrial-Grade Solvers. Beyond IPOPT, we will include comparisons with other widely-used nonlinear solvers such as SNOPT and KNITRO in the revised manuscript to provide a more comprehensive benchmark against industrial-grade alternatives.

Comment 3.3 Scaling Evidence: More rigorous scaling studies (time-to-solution vs. problem dimension, number of constraints) are needed to conclusively demonstrate the "order of magnitude" speedup claim.

Response: We thank the reviewer for this suggestion regarding more rigorous scaling studies. We agree that comprehensive scaling analysis is important to substantiate our efficiency claims. In the revised manuscript, we will include additional experiments that systematically evaluate time-to-solution versus problem dimension and number of constraints.

Specifically, we conduct

Question 1. Practical Verification of Assumption 2: How can the strong requirements on uniform Jacobian and function approximation errors be measured or guaranteed in practice, given that the true homeomorphism is unknown?

Response: We thank the reviewer for this follow-up question regarding the practical justification of Assumption 2. We refer the reviewer to our detailed response to Comment 2 above, where we provide comprehensive theoretical and practical justifications for both conditions.

To briefly summarize the practical verification aspects: (i) the function approximation error can be directly verified by measuring the Hausdorff distance $d_H(\Phi(\mathcal{B}), \mathcal{K})$ through sampling, and our experiments in Appendix G.1 confirm that trained INNs consistently achieve small Hausdorff distances; (ii) while the Jacobian approximation error is more challenging to verify directly, Appendix E.1 establishes a theoretical connection showing that small Hausdorff distance implies the existence of a target homeomorphism with controlled Jacobian deviation via optimal transport theory. Together, these results bridge our theoretical analysis with practical performance by demonstrating that the conditions in Assumption 2 are achievable through our INN training framework and can be partially monitored through empirical metrics during training.

Despite the relatively strong nature of Assumption 2, we emphasize that our work advances the state-of-the-art

by providing the first convergence analysis for homeomorphism-based reparameterization methods in the non-convex constrained optimization setting with learned mappings, which is a non-trivial and practically relevant extension.

Question 2. Novelty Distinction: Beyond integrating prior ideas of homeomorphic projection and reparameterization, what is the fundamental new contribution? Is it primarily the convergence analysis for an iterative optimizer using an INN?

Response: We thank the reviewer for this question regarding the fundamental contributions of our work. We refer the reviewer to our detailed response to Comment 2 above for a comprehensive discussion of both algorithmic and theoretical distinctions from prior literature [1]–[3].

To briefly summarize, our key novelties extend beyond simply integrating prior ideas:

1. Algorithmic Contribution. While [1, 2] use INN-based projections as single-step feasibility layers for neural network outputs, we integrate learned homeomorphisms into a *complete iterative optimization framework*. This requires addressing challenges such as error accumulation across iterations and ensuring stability of the optimization dynamics under inexact projections—issues that do not arise in single-step settings.

2. Theoretical Contribution. Our convergence analysis is indeed a fundamental new contribution. We provide the first convergence guarantees for homeomorphism-based reparameterization methods in the non-convex setting with learned (approximate) mappings. Unlike [3], which assumes exact homeomorphisms for convex constraints, our analysis explicitly quantifies how INN approximation errors impact both feasibility and optimality, achieving improved $O(\epsilon^{-2})$ rates compared to existing $O(\epsilon^{-3})$ or $O(\epsilon^{-4})$ rates for comparable methods.

In essence, the novelty lies not only in the convergence analysis itself, but in establishing a principled theoretical framework that bridges learned geometric transformations with rigorous optimization guarantees in the non-convex setting.

Question 3. Experimental Breadth and Scaling: Could the evaluation be strengthened with tests on more machine-learning-relevant constrained problems and more comprehensive time-to-solution scaling analyses against a wider range of solvers?

Response: We thank the reviewer for this question regarding experimental evaluation. We refer the reviewer to our detailed responses to Comments 3.1, 3.2 and 3.3, where we address both the experimental breadth and scaling analysis concerns.

To briefly summarize: (i) for machine-learning-relevant problems, we will include additional experiments on non-uniform adversarial attacks on neural networks, which represents a practically important problem class where our method's amortization benefits are particularly valuable; (ii) for solver comparisons, we will add benchmarks against additional industrial-grade solvers including BARON, SNOPT, and KNITRO beyond the existing IPOPT comparisons; (iii) for scaling analysis, we ... .